

# Civic Match

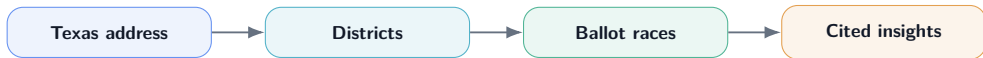
Technical Architecture for a sourced voter-information tool

Address to ballot

No source, no claim

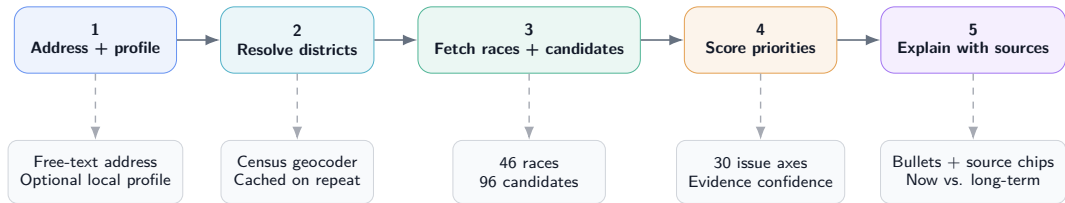
Precompute first

LLM only on fallback



Product claim: every voter-facing factual claim should trace to an explicit source URL, and missing evidence should display as unknown rather than guessed.

# Product Loop

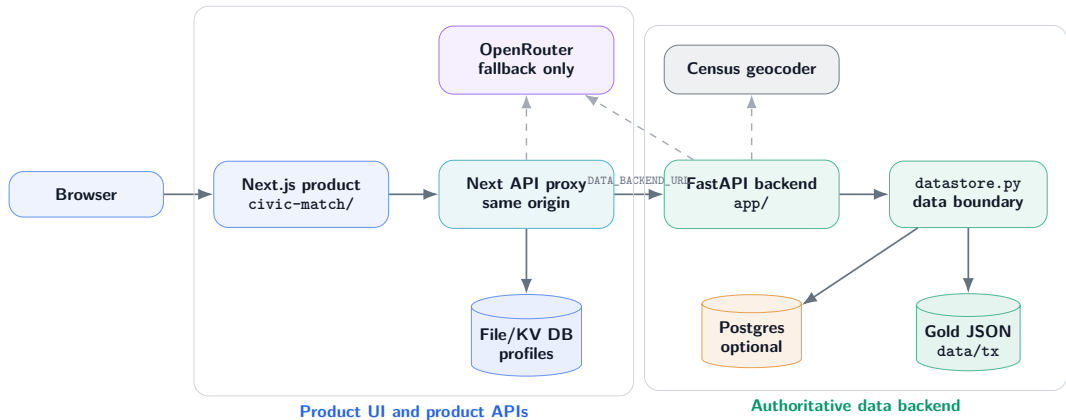


Neutral comparison, not endorsement

Missing data is a displayed measurement gap

Runtime hot path stays mostly cache reads

# Two-Service Monorepo



# Live Codebase Snapshot

Python  
**44**  
18.4k lines

TypeScript  
**50**  
6.2k lines

TSX  
**18**  
6.8k lines

FastAPI  
**3**  
root endpoints

Next API  
**15**  
route files

Pages  
**8**  
product surfaces

| Area      | Main files inspected                                          |
|-----------|---------------------------------------------------------------|
| Backend   | main.py, datastore.py, insights.py, middleware.py             |
| Pipeline  | dataset build, Postgres load, horizons, validators            |
| Frontend  | App Router pages, scoring, store, agents, data backend bridge |
| Contracts | schema, frontend schemas, demo play-book                      |

Repository root = data backend

civic-match/ = product frontend

# Gold Data Snapshot

Races

**46**

7 state + 39 federal

Candidates

**96**

embedded in ballots

Vote records

**290**

House Clerk XML

Insight files

**46**

cached by race

Graph

**157**

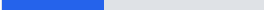
nodes / 287 edges

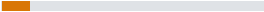
Profiles

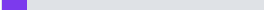
**96**

frontend records

FEC/TEC finance coverage  **79 / 96**

Roll-call vote coverage  **37 / 96**

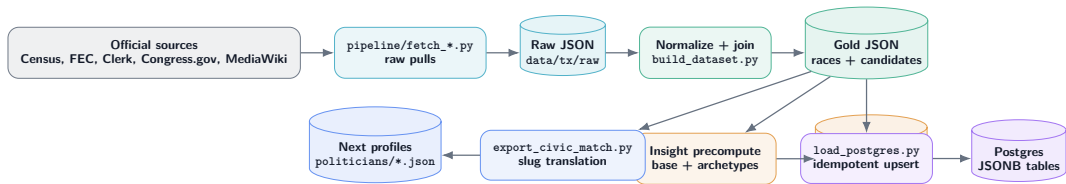
Position records present  **10 / 96**

Frontend complete profiles  **9 / 96**

Cached base insights  **46 / 46**

Expanded reference layer in `data/politicians/`: 537 Congress members, 7,453 state legislators, and 76 Texas municipal officials.

# Data Pipeline: Bronze to Runtime

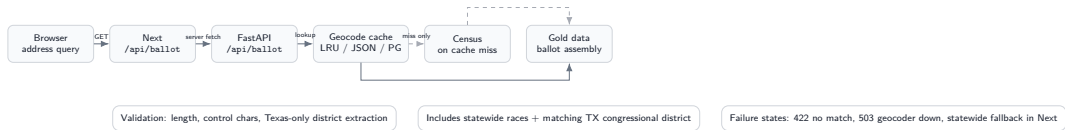


Gold files are source of truth

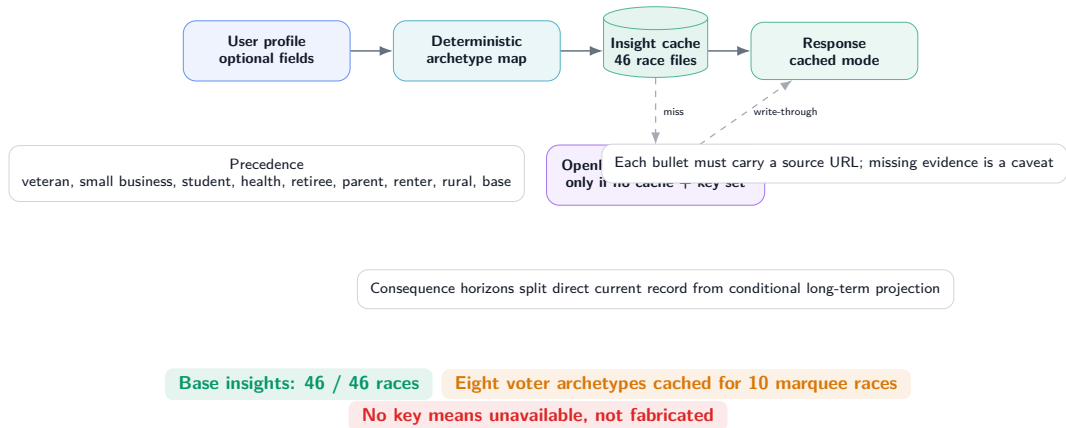
Runtime selects cached insight blocks

Postgres is production path, JSON fallback stays functional

# Ballot Hot Path

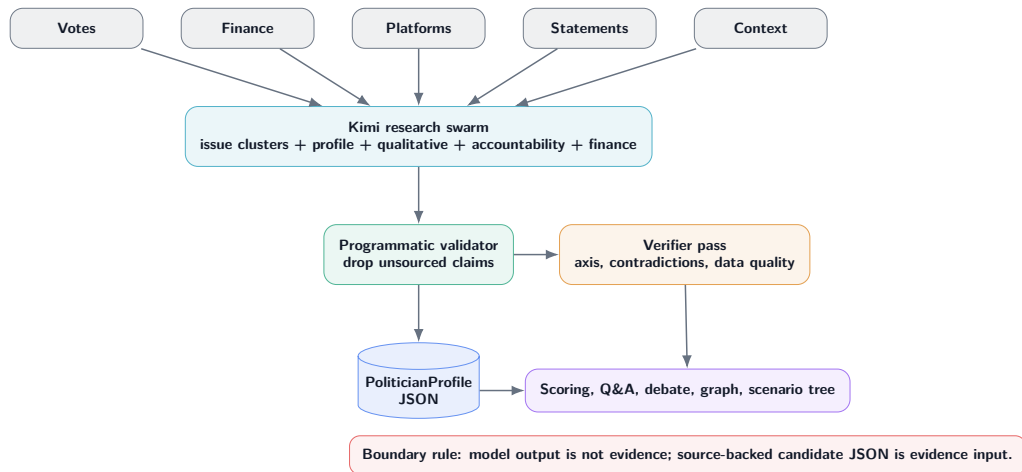


# Insight Safety Boundary





# Evidence Machine



# Scoring: Auditable Math

## Issue alignment

$$\sum w_i \times a_i \times c_i \times r_i \times q_i$$

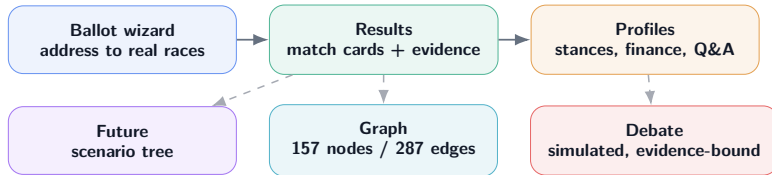
- $w_i$  user's normalized priority weight
- $a_i$  scalar closeness:  $1 - |user - candidate|$
- $c_i$  evidence confidence
- $r_i$  recency adjustment
- $q_i$  source quality multiplier

Unknown evidence lowers confidence, not alignment



UI should always expose decomposition: issue-by-issue points, top agreements, top conflicts, unknowns, and qualitative record dimensions.

# Product Surface Map



Visual risk: projections must

Visual risk: deterministic and

Visual risk: never present generated debate as actual candidate speech.

# API and Data Boundaries

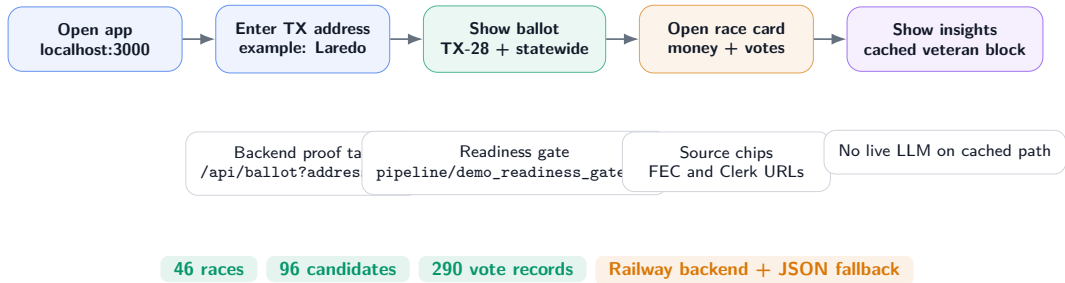
| Boundary              | Endpoint or module  | Presentation-safe claim                                                           |
|-----------------------|---------------------|-----------------------------------------------------------------------------------|
| <b>Backend health</b> | GET /healthz        | Reports data-loaded status plus race, candidate, and insight counts.              |
| <b>Ballot lookup</b>  | GET /api/ballot     | Texas address resolves to districts, then statewide plus district-specific races. |
| <b>Insights</b>       | POST /api/insights  | Picks cached archetype block before any live model fallback.                      |
| <b>Next proxy</b>     | app/api/ballot      | Browser talks same-origin Next; backend URL stays server-side.                    |
| <b>Voter insights</b> | /api/voter-insights | Translates frontend profile shape to backend archetype contract.                  |
| <b>Research</b>       | research + check    | SSE research flow for insufficient candidate data.                                |
| <b>Storage</b>        | datastore + store   | Backend uses Postgres-or-JSON; frontend uses KV Postgres-or-file DB.              |

HTTP handlers do not own storage details

Data access has explicit fallback paths

Addresses and profiles are request-local, not stored as user accounts

# Demo Path in Three Minutes



# What We Can Defend to Judges

## Accuracy is the product

Every factual claim should map to a source URL.

Unknowns lower confidence and show as gaps.

Projection language is separated from observed facts.

## Engineering posture

Precompute source-heavy work before the demo.

Keep runtime fast: cache reads plus geocoder on miss.

Postgres for production, JSON fallback for resilience.

## One-line close

Civic Match turns a Texas address into a sourced ballot explanation, while making the evidence boundary visible at every step.

# Appendix: Current Route Inventory

| Surface                                    | Method | Purpose                                                              |
|--------------------------------------------|--------|----------------------------------------------------------------------|
| <b>FastAPI</b> /healthz                    | GET    | Healthcheck and data counts.                                         |
| <b>FastAPI</b><br>/api/ballot              | GET    | Address validation, geocoding, district extraction, ballot assembly. |
| <b>FastAPI</b><br>/api/insights            | POST   | Archetype mapping, cached insights, optional OpenRouter fallback.    |
| <b>Next</b> /api/ballot                    | GET    | Same-origin proxy with statewide fallback.                           |
| <b>Next</b> /api/match                     | POST   | Pure scoring over cached profiles.                                   |
| <b>Next</b> /api/research                  | POST   | SSE candidate research flow.                                         |
| <b>Next</b><br>check-research              | GET    | Insufficient-data detection for one-click research.                  |
| <b>Next</b> Q&A / explain<br>/ debate      | POST   | Evidence-bounded generated product surfaces.                         |
| <b>Next</b> graph / sce-<br>nario / stakes | GET    | Visual civic context layers.                                         |